

Submission MICRO1-PS2-SUB01

Question MICRO1-PS2-Q01

scratch: show step

1. $33 - 4P = 7 + P \Rightarrow P^* = (33 - 7) / (4 + 1) = 5.20$

2. $Q^* = 33 - 4(5.20) = 12.20$

3. substitution gives $Q_d = Q_s$

=> FINAL: $P^* = 5.20$, $Q^* = 12.20$ (check units/interpretation)

Question MICRO1-PS2-Q02

scratch: show step

1. $\% \Delta Q = -3 / 28.5$

2. $\% \Delta P = 0.0952$

3. $E = -1.11$; $|E| > 1$ (elastic)

=> FINAL: elasticity = -1.11; elastic (check units/interpretation)

Question MICRO1-PS2-Q03

scratch: show step

1. Shift the wedge between buyer and seller prices by t.

2. Quantity falls relative to equilibrium.

3. The lost mutually beneficial trades create a positive deadweight-loss triangle.

=> FINAL: buyer price minus seller price equals tax; quantity falls; deadweight loss increases (check units/interpretation)

Question MICRO1-PS2-Q04

scratch: show step

1. $SMC = PMC + \text{marginal external cost}$, so SMC lies above PMC.

2. A Pigouvian tax equal to marginal external cost aligns private and social incentives.

=> FINAL: social MC above private MC; Pigouvian tax can internalize the external cost (check units/interpretation)

Question MICRO1-PS2-Q05

scratch: show step

1. At the ceiling, Q_d exceeds Q_s .

2. Shortage = $Q_d - Q_s$; queues or rationing allocate the scarce quantity.

=> FINAL: $Q_d > Q_s$ creates a shortage; queues/rationing may allocate units (check units/interpretation)

Question MICRO1-PS2-Q06

scratch: show step

1. $41 - 2P = 7 + P \Rightarrow P^* = (41 - 7) / (2 + 1) = 11.33$

2. $Q^* = 41 - 2(11.33) = 18.33$

3. substitution gives $Q_d = Q_s$

=> FINAL: $P^* = 11.33$, $Q^* = 18.33$ (check units/interpretation)

Question MICRO1-PS2-Q07

scratch: show step

1. $\% \Delta Q = -4/38.0$

2. $\% \Delta P = 0.0952$

3. $E = -1.11$; $|E| > 1$ (elastic)

=> FINAL: elasticity = -1.11; elastic (check units/interpretation)

Question MICRO1-PS2-Q08

scratch: show step

1. Shift the wedge between buyer and seller prices by t .

2. Quantity falls relative to equilibrium.

3. The lost mutually beneficial trades create a positive deadweight-loss triangle.

=> FINAL: buyer price minus seller price equals tax; quantity falls; deadweight loss increases (check units/interpretation)